

APPLIED DATA ANALYSIS

Faculty of National Economy
Bratislava University of Economics and Business
Department of Economic Policy

Instructors: Tomáš Oleš, Martin Vaško, Ella Sargsyan
Email: tomas.oles@euba.sk; martin.vasko@euba.sk;
esargsyan@cerge-ei-foundation.org
Office Hours: Wednesday: 1–3:00 PM
Office Location: Old Building, 4B40
Credit Hours: 6
Term: Fall 2026

COURSE DESCRIPTION

The first goal of the course is to teach students knowledge and skills in modern methods of applied data analysis and statistical learning, including the use of R software, to conduct empirical economic research and design research approaches and methods for solving economic problems.

The second aim of this course is to acquaint students with basic econometric concepts and methods which are necessary to read and understand current empirical research in economics, as well as plan and execute independent research projects. The course will introduce the idea of regression analysis and cover some of the most recent econometric techniques central to modern econometric practice, which aims to identify causal relationships.

The course is delivered in two parts. **Part I – Applied Data Analysis in R** (Weeks 1–7) covers the data-analytic toolkit: the R language, data visualization, data wrangling, text analysis, and an introduction to statistical learning through classification and clustering. **Part II – Econometrics in a Nutshell** (Weeks 8–13) is delivered as a self-contained module by Ella Sargsyan, Ph.D. (CERGE-EI Foundation), and covers regression analysis and modern causal-inference techniques.

STUDENT LEARNING OUTCOMES

Upon completing the course, students should acquire knowledge of modern methods of data research and visualization, classification, clustering, linear regression, and general data analysis. Skills in working with data, which they will be able to apply in their own empirical research. They will also gain advanced skills in using modern software (R) in empirical economic research and will be able to write their own functions.

Students will know basic econometric concepts, basic estimation methods, and methods for testing statistical hypotheses. They will be able to apply standard methods of constructing econometric models, process statistical information, obtain statistically sound conclusions, and give meaningful interpretation to the results of the estimated econometric models. In addition, students will gain real data processing skills, using econometric packages for building and estimating regression models in R.

PREREQUISITES

Introduction to Statistics.

TEXTBOOKS AND COURSE MATERIAL

- Wickham, H., Çetinkaya-Rundel, M., Grolemund, G. (2023). *R for Data Science*. O'Reilly Media, Inc. Freely available at <https://r4ds.hadley.nz/>.

- Gareth, J., Daniela, W., Trevor, H., Robert, T. (2013). *An Introduction to Statistical Learning: With Applications in R*. Springer. Freely available at <https://www.statlearning.com/>.
- Silge, J., Robinson, D. (2017). *Text Mining with R: A Tidy Approach*. O'Reilly Media, Inc. Freely available at <https://www.tidytextmining.com/>.
- Wooldridge, Jeffrey M. *Introductory Econometrics: A Modern Approach*. 5th ed., US, 2012.
- Angrist, J. D., Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Tutorial on basic R: https://cran.r-project.org/doc/contrib/Paradis-rdebuts_en.pdf.

All course material (relevant textbook chapters, reading papers and lecture slides) will be uploaded on the Moodle course webpage.

GRADING AND EVALUATION

The final grade is made up of three components.

- **Part I — quizzes and home assignments: 30%.** Short quizzes and assignments given during and after the lectures of Part I, together with the two home assignments based on R.
- **Part II — Econometrics in a Nutshell: 35%.** Assessed within that module.
- **Final data project: 35%.** A project on a question and a dataset of the student's own choosing, presented at the end of the semester. It draws on both Part I and Part II and is the concluding piece of work for the course.

In-class questions and discussions are not graded, but active participation may be taken into account at the instructor's discretion.

Final grade calculation: there is no final written exam. Students do not need to reach a given threshold in each individual component; the final grade is based on the cumulative score.

IMPORTANT DATES

- **Part I (Applied Data Analysis in R):** September 22 – November 4, 2026
- **Part II (Econometrics in a Nutshell):** October 26 – December 4, 2026
- **Final project presentations:** December 7 – 11, 2026
- **Make-up week:** December 14 – 18, 2026

Class times.

	Part I — Applied Data Analysis in R	Part II — Econometrics in a Nutshell
Course code	NND21254/21 (English class)	CERGE-EI module
Day	Wednesday, from 23 Sep 2026	Lecture: Mon + Thu, from 26 Oct 2026 Exercise: Wednesday
Lecture	11:00–12:30, room 4B26	11:00–12:30 CET, online
Exercise	13:30–15:00, room D203	13:30–15:00, room D203 (in person)

This syllabus states the times for the **English class**. The Slovak class (NND21210/21) meets on Tuesdays — see the Slovak version of this syllabus.

In Part II the lectures are delivered **online** by Ella Sargsyan, while the exercise session remains **in person** in room D203 at the usual Wednesday time.

The instructor reserves the right to change the content of the course material if he perceives a need due to postponement of class caused by inclement weather, instructor illness, etc., or due to the pace of the course.

ACADEMIC INTEGRITY

Academic integrity is essential to the pursuit of learning and scholarship in a university, and to ensuring that a degree from the Bratislava University of Economics and Business is a strong signal of each student's individual academic achievement. As a result, the University treats cases of cheating and plagiarism very seriously. The Bratislava University of Economics and Business Ethical Code (<https://euba.sk/univerzita/eticky-kodex>) outlines the behaviors that constitute academic dishonesty and the processes for addressing academic offenses. Potential offenses include, but are not limited to:

In papers and assignments: Using someone else's ideas or words without appropriate acknowledgment; submitting your own work in more than one course without the permission of the instructor; making up sources or facts; obtaining or providing unauthorized assistance on any assignment.

On tests and exams: Using or possessing unauthorized aids; looking at someone else's answers during an exam or test; misrepresenting your identity / pretending to be someone else.

In academic work: Falsifying documents or grades; falsifying or altering any documentation required by the University, including (but not limited to) medical forms.

All suspected cases of academic dishonesty will be investigated following procedures outlined in the Ethical Code. If students have questions or concerns about what constitutes appropriate academic behavior or appropriate research and citation methods, they are expected to seek out additional information on academic integrity from their instructors or from other institutional resources.

CLASS SCHEDULE

Wk	Topic Description	Dates & Instructor
PART I — APPLIED DATA ANALYSIS IN R		
1	R basics. Basic R syntax, data types, vectors arithmetic, indexing, sorting, sorting using <code>dplyr</code> , and plotting using basic packages. GitHub. <i>Reading: Gareth et al. (2013), Ch. 1. Wickham et al. (2023), Ch. 1, 2, 4–6.</i>	Wed 23 Sep 2026 Lecture 11:00–12:30, 4B26 Exercise 13:30–15:00, D203 <i>T. Oleš</i>
2	Visualization Part 1. Data visualization principles, creating custom plots with <code>ggplot2</code> , and studying the advantages and pitfalls of widely-used plots. <i>Reading: Wickham et al. (2023), Ch. 3, 7.</i>	Wed 30 Sep 2026 Lecture 11:00–12:30, 4B26 Exercise 13:30–15:00, D203 <i>T. Oleš</i>
3	Visualization Part 2. Spatial visualization principles, creating custom plots with <code>tmap</code> . Spatial data overview (i.e. points, lines and polygons) and ending with a brief raster example. <i>Reading: Wickham et al. (2023), Ch. 3, 7.</i>	Wed 7 Oct 2026 Lecture 11:00–12:30, 4B26 Exercise 13:30–15:00, D203 <i>T. Oleš</i>
4	Data Summarizing, Tidying and Joining. Importing data from different file formats, web scraping, tidy data with <code>tidyverse</code> , wrangling data with <code>dplyr</code> , handling date and time formats. <i>Reading: Wickham et al. (2023), Ch. 9–16.</i>	Wed 14 Oct 2026 Lecture 11:00–12:30, 4B26 Exercise 13:30–15:00, D203 <i>T. Oleš</i>
5	Text Analysis. The tidy text format and tokenization. Stop words and word frequencies. Sentiment analysis with tidy data (Bing, AFINN, NRC lexicons). Term frequency, inverse document frequency and <code>tf-idf</code> . <i>Reading: Silge and Robinson (2017), Ch. 1–3.</i>	Wed 21 Oct 2026 Lecture 11:00–12:30, 4B26 Exercise 13:30–15:00, D203 <i>T. Oleš</i>
6–7**	Classification and Clustering.** Introduction to qualitative classification. Classifiers: logistic regression, naive Bayes, K-nearest neighbors, generalized linear models. Overfitting, confusion matrix, specificity and sensitivity. Unsupervised learning and the notion of similarity: K-means, hierarchical clustering and dendrograms, choosing the number of clusters, principal component analysis and its use in cluster analysis. <i>Reading: Gareth et al. (2013), Ch. 4 and Ch. 12.</i>	Wed 28 Oct and 4 Nov 2026 Lecture 11:00–12:30, 4B26 Exercise 13:30–15:00, D203 <i>T. Oleš, M. Vaško</i>
PART II — ECONOMETRICS IN A NUTSHELL		

Wk	Topic Description	Dates & Instructor
8	The Nature of Econometrics and Economic Data. What is Econometrics and why do we need it? Steps in empirical economic analysis. The structure of economics data. Causality and the notion of “ceteris paribus” in econometric data. Definition of simple regression model. <i>Reading: Wooldridge (2012), Ch. 1; Ch. 2 (2.1).</i>	Lecture (online) Mon 26 Oct 11:00–12:30 Thu 29 Oct 11:00–12:30 Exercise (in person) Wed 28 Oct 13:30–15:00, D203 <i>E. Sargsyan</i>
9	Regression Analysis with Cross-Sectional Data: Part 1. Ordinary Least Squares. Properties of Ordinary Least Squares. Assumptions of the simple regression model. <i>Reading: Wooldridge (2012), Ch. 2 (2.2–2.5).</i>	Lecture (online) Mon 2 Nov 11:00–12:30 Thu 5 Nov 11:00–12:30 Exercise (in person) Wed 4 Nov 13:30–15:00, D203 <i>E. Sargsyan</i>
10	Regression Analysis with Cross-Sectional Data: Part 2. Multiple regression analysis. Hypothesis testing. Reporting regression results. <i>Reading: Wooldridge (2012), Ch. 3; Ch. 4 (4.2–4.6).</i>	Lecture (online) Mon 9 Nov 11:00–12:30 Thu 12 Nov 11:00–12:30 Exercise (in person) Wed 11 Nov 13:30–15:00, D203 <i>E. Sargsyan</i>
11	Regression Analysis: Further Topics and Issues. Units of measurement and functional forms. Regression analysis with qualitative information. Heteroskedasticity and serial correlation. Endogeneity: omitted variables, selection, and reverse causality. <i>Reading: Wooldridge (2012), Ch. 2 (2.4); Ch. 3 (3.3); Ch. 7 (7.1–7.4); Ch. 8.</i>	Lecture (online) Mon 16 Nov 11:00–12:30 Thu 19 Nov 11:00–12:30 Exercise (in person) Wed 18 Nov 13:30–15:00, D203 <i>E. Sargsyan</i>
12	Solving Endogeneity. Instrumental Variables. Regression Discontinuity Design. <i>Reading: Wooldridge (2012), Ch. 15 (15.1–15.3); Angrist and Pischke (2009), Ch. 6 (pp. 189–195).</i> <i>Papers to be discussed: Acemoglu, D., & Johnson, S. (2005). “Unbundling institutions.” Journal of Political Economy, 113(5), 949–995. Carpenter, C., & Dobkin, C. (2009). “The effect of alcohol consumption on mortality: regression discontinuity evidence from the minimum drinking age.” American Economic Journal: Applied Economics, 1(1), 164–182.</i>	Lecture (online) Mon 23 Nov 11:00–12:30 Thu 26 Nov 11:00–12:30 Exercise (in person) Wed 25 Nov 13:30–15:00, D203 <i>E. Sargsyan</i>
13	Panel Data Methods. Introducing time dimension. Fixed effects model. Differences-in-Differences. <i>Reading: Wooldridge (2012), Ch. 14 (14.1); Angrist and Pischke (2009), Ch. 5 (pp. 169–182).</i> <i>Paper to be discussed: Card, D., & Krueger, A. B. (1994). “Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania.” The American Economic Review, 84(4), 772–793.</i>	Lecture (online) Mon 30 Nov 11:00–12:30 Thu 3 Dec 11:00–12:30 Exercise (in person) Wed 2 Dec 13:30–15:00, D203 <i>E. Sargsyan</i>

** Covered based on the strength of the group. These two sessions run in parallel with the opening weeks of Part II.

“We are what we repeatedly do. Excellence, then, is not an act, but a habit.” — Aristotle